

Lab # 4

Introduction to vi editor, File links and Local user management

A. Introduction to vi Editor:

A text editor is a program used to edit files that are composed of text: a letter, C program, or a system configuration file. While there are many such editors available for Linux, the only editor that you are guaranteed to find on any UNIX or Linux system is vi-- the “visual editor”. Vi is not the easiest editor to use, nor is it very self-explanatory. However, because vi is so common in the UNIX/Linux world, and sometimes necessary, it deserves discussion here.

- i) Opening file vi Editor
- ii) Vi Commands
- iii) Moving within file (Navigation)
- iv) Control Commands (Scrolling)
- v) Inserting and replacing text
- vi) Deleting Characters lines
- vii) Copy and Paste
- viii) Save and Exit
- ix) Search and Replace
- x) Block commands (Copy, Moving and Deleting)

- i) Opening file vi Editor
\$ vi <filename>
Example: \$ vi text.txt

<https://www.geeksforgeeks.org/vi-editor-unix/#how-to-open-vi-editor>

\$ vi <filename>

\$ vi text.txt

Command mode or Normal mode

Press i--Insert mode

I am in BSCS

Press Esc--Normal Mode again

Press i--Insert Mode-- Editing only

I am in SSUET.

I am student in ssuet.

I am in fifth semester.

Error- press esc

Only save- press : colon and w –write

: q --for exit

: q ! --for exit without changes

: wq –write,save and then quit dd -- delete one line

u – undo

yy – copy one line in command mode.

p – paste

y1 – two lines copy

y3 – three lines copy

d2 – three lines delete

cc – cut

In command mode -- /ssuet for search

\$ vimtutor

II. File links:

a) Symbolic link:

Just like shortcuts in windows. File links are same as this. Create link of the file.
\$ echo “a test file” > file1.txt

\$ cat file.txt

\$ ls –lhtr

\$ vi file1.txt

\$ vi latest-file.txt

\$ ln –s file1.txt latest-file

- This will create latest-file symbolic link. It is of 9 byte in size. This will point to file1.txt

\$ cat file1.txt

```
$ cat latest-file
```

```
$ vi latest-file
```

- The changes in latest-file will change in file1.txt

```
$ cat file1.txt
```

- Also we can create symbolic link of directory.

```
$ ln -s mydir/bilal
```

```
$ls -lhtr
```

```
$ cd bilal
```

```
$ ls
```

- It will show the content in original directory.

```
$ cd ..
```

```
$ rm bilal
```

- The original directory will remain save.

```
$ rm file1.txt
```

- After this symbolic link file not work latest-file.

```
$ rm latest-file
```

b) Hard Link:

```
$ echo "My test file" > file1.txt
```

```
$ ls -lhtr
```

\$ ln file1.txt latest-data.txt

This create the hard link. Hard link will same in size that of original file.

- This will show in regular file format.

```
$ vi file1.txt
```

- If we make changes in original file it will take place in hard link also.

```
$ vi latest-data.txt
```

- If we change in hard link it will also take place in original file.

```
$ rm file1.txt
```

- If we remove original file, hard link will remain there.

```
$ cat latest-data.txt
```

- Hard link is the copy of original file.

```
$ rm latest-data.txt
```

III. User Management:

```
$ which find /usr/bin/find
```

Which command show that where is the file in the directory.

```
$ which ls
```

```
/bin/ls
```

```
$ whereis find
```

In more detail.

Shows complete location of the program.

```
$ who
```

Shows how many users are logged in.

```
$ w
```

Shows users logged in in more details.

```
$ whoami
```

Tell which user logged in

```
$ last
```

Tell which user logged in last time.

```
$ su -
```

Switch user

```
$ su - root
```

If there is only - sign then it is assumed to login as root user.

\$ su – bilal

To login as bilal user.

su – bilal

If you are login as root user and want to switch to general user. No password required.

TASK

- 1.Create a MyInfo.txt using vi editor.
- 2.Write Your name , roll number and section in MyInfo.txt
- 3.Save the File.
- 4.Delete your section from MyInfo.txt.
- 5.Show content of MyInfo.txt using cat.
- 6.Print host name
- 7.Print your(system) user name.
- 8.Show the locatoin where ls ,pwd ,rmdir & cat source code is present.